

Title

Curcumin supplementation slows early renal biomarker progression in feline stage 3 chronic kidney disease

Abstract

Objective: To assess whether curcumin supplementation provides renal protective benefits beyond standard medical management. **Setting:** Veterinary nephrology clinic. **Subjects:** 16 cats with IRIS stage 3 chronic kidney disease randomly assigned to medical management plus curcumin (n=8) or medical management alone (n=8). **Interventions:** 80 mg curcumin daily in divided doses for 60 days. **Outcomes:** Baseline renal parameters were equivalent between groups. The primary endpoint, serum symmetric dimethylarginine, declined more substantially in the curcumin-supplemented group compared to standard treatment alone at days 30, 45, and 60 ($p < 0.01$ for all timepoints). Urine protein-to-creatinine ratio also showed greater improvement over time in the supplemented group, with significant between-group differences at later study visits. Antioxidant capacity increased while oxidative stress markers decreased in the curcumin group. Inflammatory markers and apoptosis markers showed trends consistent with reduced kidney cellular injury. All cats tolerated supplementation well without adverse events. **Conclusions:** Curcumin co-treatment appeared to slow renal biomarker deterioration in cats with stage 3 chronic kidney disease.